A one-parameter crack detector, and what its scores do and do not mean

DRAFT — not submitted. Every number is generated from measurement files by paper/make_paper.py; nothing in this document is typed by hand twice.

Abstract

We describe a crack segmentation detector with a single free parameter and no learned weights, and measure it on two public crack datasets with pixel-exact ground truth. On the DeepCrack test split it reaches mean per-image F1 0.7932 and IoU 0.6809, ranking first of seven among classical detectors tuned identically on the same images; on CrackForest it reaches F1 0.4498 and IoU 0.3070, ranking third. One constant refitted on one dataset carries to the other. We then show that the scores themselves are the fragile part: a reconstruction of the label that is exactly right about where the crack is and wrong only about how wide it is loses most of its IoU, and the shape of that loss differs between corpora. We report the measurement practices this forced on us, and several negative results, including one that refutes the explanation we ourselves offered first.

Status of this draft. Section 7 lists what is missing before submission.

1 Why another crack detector

There are many. Most are learned, most report F1 or IoU against a benchmark, and most are compared against each other at their own authors' operating points. Our interest is narrow and practical: a materials laboratory needs crack measurements out of scanning-electron and transmission X-ray micrographs, on a machine with no GPU, from a tool a researcher can audit line by line and correct by hand. That rules out most of the field, and it makes the natural baseline not a network but a threshold.

The detector here is close to that baseline. It flattens the background at seven scales, takes pixels that are dark at any of them, applies a robust outlier cut and an absolute darkness floor, and cleans the result. It has one free constant, mad_k. There is no training step and no model file.

Reporting such a thing honestly turns out to be harder than building it.

2 The detector

Figure 1 is the whole path. Two gates do the detection: a robust median-absolute-deviation cut at mad_k standard deviations, and an absolute floor on intensity. Everything else is preparation and cleanup.

Two decisions are worth stating because both were made by measurement and both went against the intuition that produced the code in the first place.

The region classifier is switched off. An eight-feature classifier scored each candidate region, and deleting it improved every metric with no free parameters: paired median IoU +0.2296 over thirteen specimens, p = 0.0024. Three specimens that had scored recall 0.0000 at every candidate setting ever tried were never invisible to the proposal stage; the classifier was rejecting them.

Tiles cannot stand in for frames. Rules validated on 512 px crops deleted 40–43% of hand-marked crack when run on whole frames. The widest crack in the training crops was 43.7 px; on full frames crack widths reach 178 px, so a region rule fitted on crops is extrapolating on the object it most needs to keep.

3 On public data with pixel-exact ground truth

Both datasets are scored at 0 px tolerance, with mad_k chosen leave-one-image-out. Image-level grouping is the right unit here — these are independent road scenes, not repeated views of one specimen.

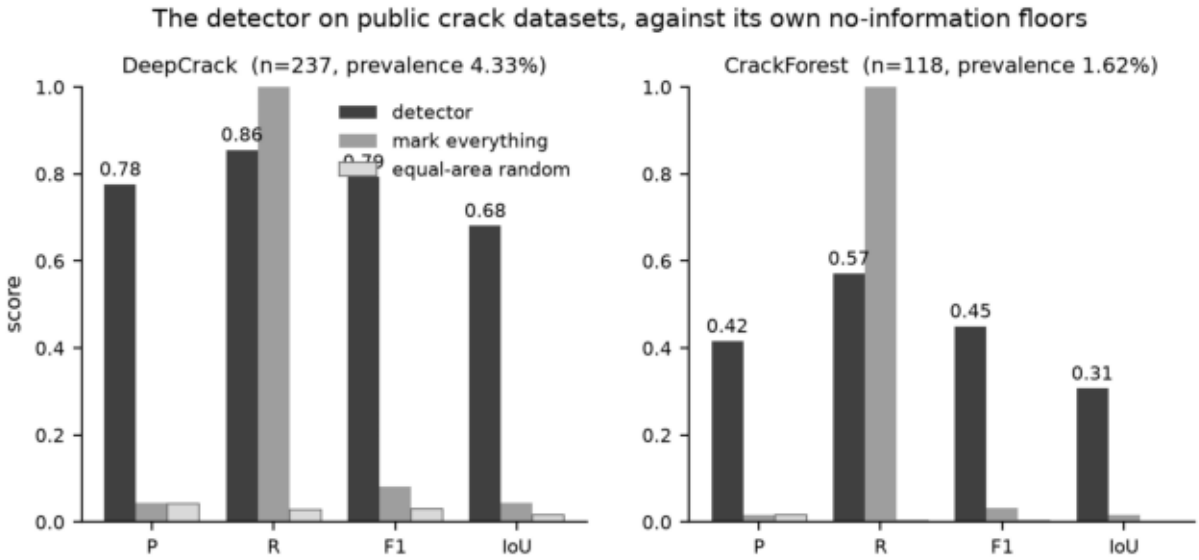


Figure 1. P1 - public scores

Against the no-information floors the margin is large: on DeepCrack the detector is 9.8 times the F1 of marking every pixel and 15.7 times its IoU; on CrackForest, 14.2 and 19.0 times. A mask of the same area with its pixels scattered at random scores 0.0311 and 0.0023.

3.1 One constant, two datasets

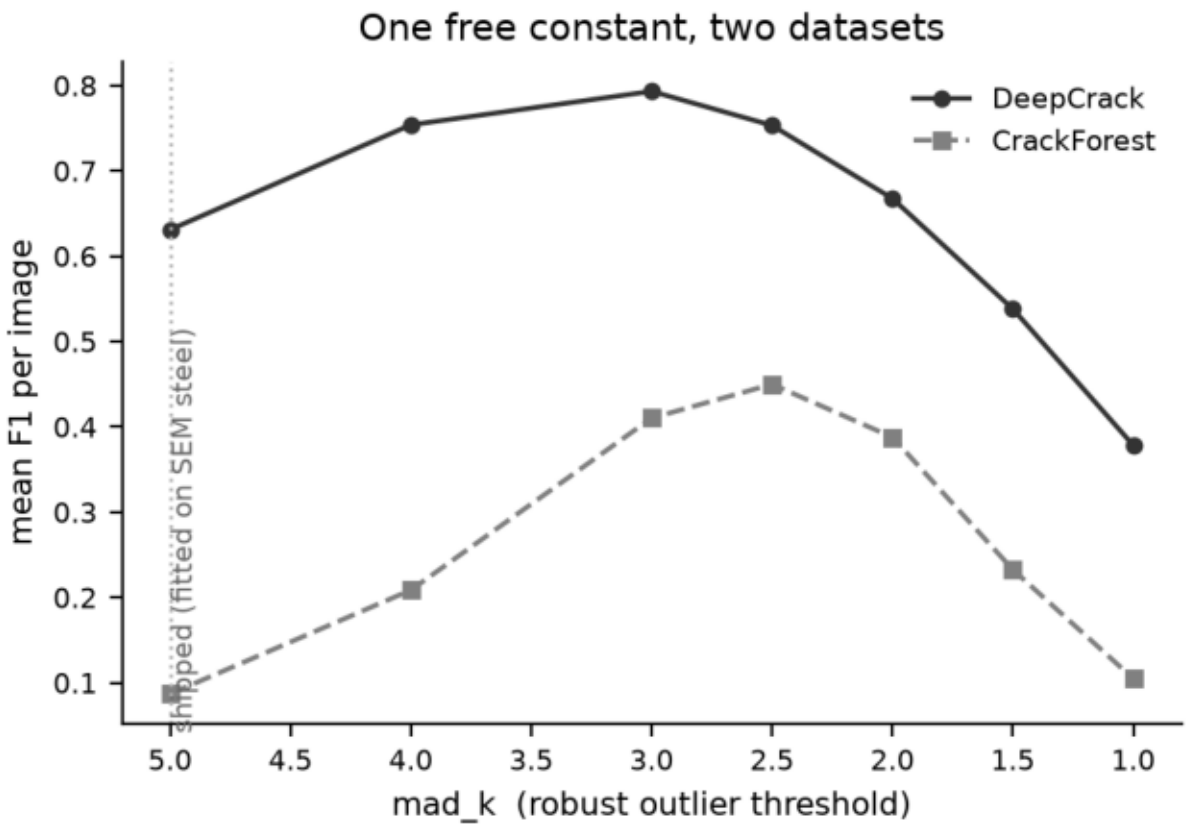


Figure 2. P2 - transfer

The shipped constant was fitted on SEM steel. Carried unchanged to road images it is poor on CrackForest (0.0868) and adequate on DeepCrack (0.6310). Refitted, each dataset peaks within one grid step of the other's optimum. That is the difference between "this needs retuning per corpus" and "this needs retuning per imaging modality", and only the second is a tolerable property for a tool.

3.2 Against classical detectors

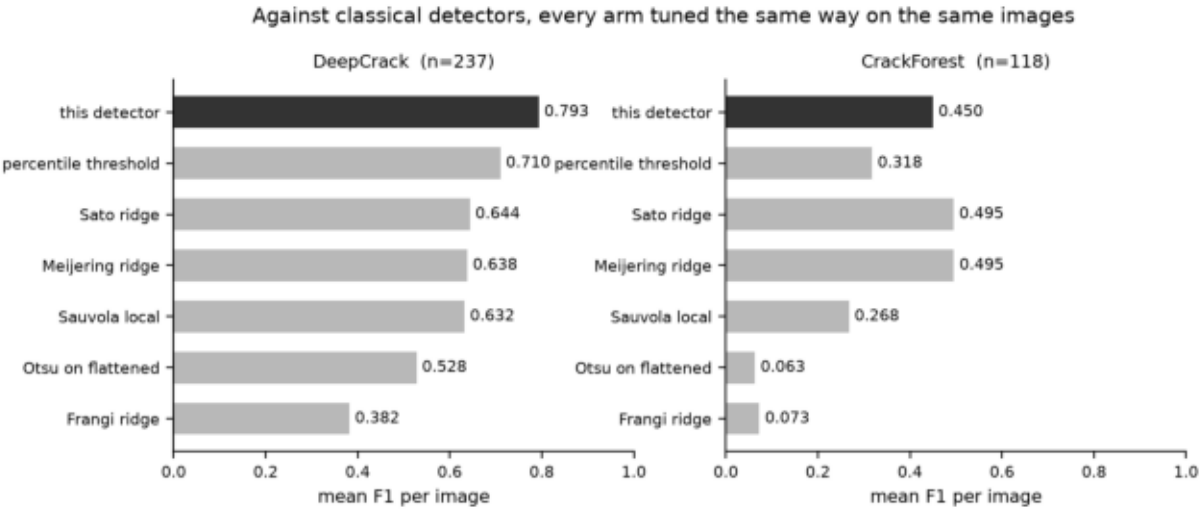


Figure 3. P4 - baselines

Every arm gets its own best parameter by the same leave-one-image-out rule, on the same images, through the same scoring function. The detector ranks first of seven on DeepCrack

and third on CrackForest, where two ridge filters beat it.

It is worth naming what this comparison is not. These are classical baselines, not the trained networks that hold the published state of the art on both datasets — about 0.86 F1 on DeepCrack and 0.7535 on CrackForest, both trained on the dataset in question. A one-parameter detector landing at 0.7932 and 0.4498 without training is a statement about how much of the benchmark is reachable without learning, not a claim to have beaten anyone.

4 What the scores measure

This is the part we would most want another group to check.

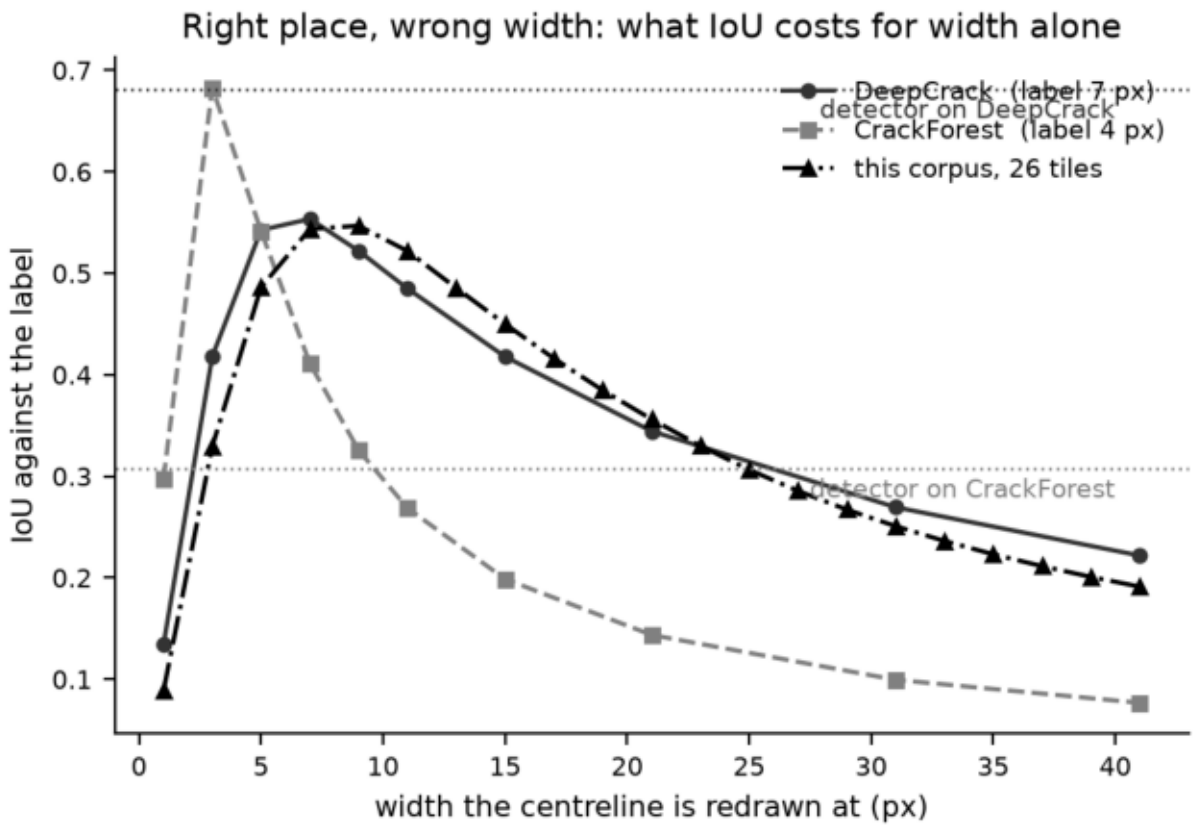


Figure 4. P3 - width sensitivity

Take the ground truth, reduce it to its centreline, and redraw it at a fixed width. The result is exactly right about where every crack is and wrong only about how wide. Scored against the label it should be nearly perfect, and it is not: on CrackForest the best such reconstruction reaches IoU 0.6827 and on DeepCrack 0.5535. Most of the available IoU is being spent on agreeing about width.

Two consequences.

A score is not comparable across corpora whose annotation differs. The curves peak at different widths and fall away at different rates, so the same detector's IoU can differ between datasets without its localisation differing at all.

But width does not explain our own gap, and we initially claimed it did. The detector scores IoU 0.6809 on DeepCrack and roughly 0.33–0.42 on our laboratory corpus, and the obvious explanation was that our hand-painted labels are much broader. Measured, that explanation fails: the constant-width curve peaks at 0.5535 on DeepCrack and 0.5467 on the corpus tiles — within 0.01 of each other. Whatever makes the laboratory corpus harder for this detector, it is not the width of the brush.

Note also that the detector exceeds the constant-width reconstruction on DeepCrack (0.6809 against 0.5535). The reconstruction is therefore not an upper bound, and we do not present it as one: a real label varies in width along its length, and a detector that tracks that variation can beat a constant-width redraw of the same centreline. We had to be told this by the measurement, having first written the figure up as a ceiling.

5 Negative results

These cost the most time and are the most likely to save someone else's.

Retuning the detector for a different modality does nothing. Applying the SEM-calibrated detector to transmission X-ray tomography, we swept its two gating constants singly and jointly, plus every morphology and size floor, under leave-one-specimen-out selection. No arm beat doing nothing on held-out data; the joint optimum lost 0.021 F1 when its selection was held out, and resolution-matching the scale ladder to the measured width ratio moved F1 by 0.0008.

That corpus cannot answer the question anyway. It holds four specimens, so a paired test has an exact two-sided floor of p = 0.125 — no result on it can reach significance at any conventional level, a win or a null. One specimen carries 57.7% of all adjudicated not-crack pixels, so the pooled score surface is essentially that specimen's.

F1 is width-gameable where the labels are broad. On the same corpus a by-construction-correct 3 px crack scores F1 0.1260, below marking every pixel (0.1499), while F1 climbs to 0.64 on marked width alone. "Beats the no-information floor on F1" is necessary and not sufficient; precision lift separates where F1 does not.

Six precision rules validated on crops were deleters on frames. See §2.

6 Measurement practices this forced

Stated as rules because each one followed a specific error of ours.

- Report the no-information floor beside every score. Marking every pixel scores F1 0.0809 on DeepCrack and 0.0316 on CrackForest. A bare score cannot be read without them.
- Choose parameters by grouped hold-out, and say what the group is. Regrouping alone moved an AUC by +0.0458 in our data, which is larger than most effects we chased.
- A difference of medians is not a paired difference. One result moved 0.0742 as medians and 0.0000 paired.
- State the resolution of the comparison. Our laboratory corpus detects about +0.14 F1 for a new method; we reported smaller differences for weeks before measuring this.
- Controls must be able to fail. Every control we built was broken on purpose to check it noticed; two did not and were rewritten.
- An oracle must match the arity of the thing it bounds. The constant-width reconstruction in §4 is not an upper bound, and we described it as one until the data said otherwise.

7 What this draft is missing

Listed rather than hidden.

- No laboratory results are reported. By choice: the specimen measurements are not being published here. Sections 2 and 5 cite effects measured on that corpus as methodological evidence; a reviewer cannot check those without the data, and they should be read as weaker than the public-dataset numbers.
- The corpus series in Figure 3 carries no stroke width. The instrument that would measure it comparably needs a tile cache not present on this machine.

- No timing, and no comparison against a trained network on equal footing. A fair speed claim needs matched hardware, and an earlier attempt found this detector 1.6x slower than a single-pass U-Net at p = 0.0001.
- Novelty is not claimed. Fifty-five framings of this work were checked against prior art and all were already owned. What is offered is measurement.

Appendix A. The detector, end to end



Figure 5. Every constant read from analysis/detector_overrides.py when the plate was drawn.